



Practice Graded Assignment: Course project

★ AI Grading

After submitting your assignment and completing your required peer reviews, you'll receive an AI-generated grade based on the assignment rubrics. You'll then have the option to have your assignment reviewed by your peers instead.

Your data will be used in accordance with [Coursera's Privacy Notice](#).

Instructions My submission Peers to review Discussions

In this project, you will demonstrate your ability to perform the following:

- Protect a database using Azure security features.

You will create two deliverables that document the steps and screenshots of activities related to securing data. These deliverables are:

- Container security project solution
- Audit logs project solution

Your detailed tasks will include the following:

- **Secure a container with a shared access signature (SAS)**
 - Create an Azure Storage account with geo-redundant storage and a container with the public access level set to Blob.
 - Upload files to Blob Storage.
 - Create SAS tokens.
 - Download and install Azure Storage Explorer.
 - Connect to the Storage account using Azure Storage Explorer.
- **Configure audit logs**
 - Create a resource group.
 - Create Azure Database for MySQL servers.
 - Create an Azure Key Vault and a user-assigned managed identity and set the proper permissions for key operations.
 - Configure customer-managed key.
 - Use data encryption for restore.
 - Configure an audit log for connection and general events.

At the end of this project, you will have your version of the following deliverables for submission:

- *Container security project solution.docx*
- *Audit logs project solution.docx*

As part of this project, you will also review the project submissions of two (2) peers using the peer review rubric provided.

You should plan to complete this course project in about thirty (30) minutes. Also, allow up to one hour to complete the graded quiz that follows this project.

^ Step-By-Step Assignment Instructions

Refreshing your knowledge

Before starting on the project or while doing the project, you can refresh your knowledge by referring to the

readings, videos, and associated exercise/solution documents in the below lessons:

- Week 1, Lesson 2: Shared access signature tokens
- Week 1, Lesson 3: Shared keys
- Week 3, Lesson 2: Auditing and monitoring
- Week 4, Lesson 1: Defender for Cloud
- Week 4, Lesson 2: Data masking and encryption

Project scenario

In this project, you will go through the typical activities that an Azure security engineer would perform in an organization.

Imagine you've recently joined a large organization as an Azure security engineer. You are responsible for acting on two problem statements:

Problem statement 1: Your company has recently partnered with a new vendor. The vendor specializes in creative work and will be responsible for creating design assets for a new ad campaign your company will launch.

The vendor needs access to your company's brand guidelines, including logos, ad campaign pictures, and sound files. The marketing department has requested your help to develop an approach where these files can be shared with the vendor, but the link should be accessible for no more than a week.

As a security engineer, you are responsible for devising an appropriate solution to address this requirement.

Task: Secure a container with SAS.

Problem statement 2: Your manager wants you to ensure audit logs are configured for a new database while implementing encryption. It should track all actions within schemas, tables, and rows and any changes to specific content. Audit logs will help ensure accountability, prevent inappropriate actions, and facilitate monitoring and data gathering for specific database activities.

You intend to configure audit logs to fulfill this requirement.

Task: Configure audit logs.

Setting up the system

This project consists of tasks you should complete on the Azure platform. Before you begin, you must have the following:

- Access to Azure to manage and configure an Azure Storage account, the Azure Blob Storage service, and associated resources.
- An application, such as Snipping Tool, to capture screenshots
- An application, such as Microsoft Word, to create the two deliverables: *Container security project solution* and *Audit logs project solution*.

Guidelines for the deliverables

We've provided two templates to help you document your work and build your deliverables. The templates provide starting points for the tasks you need to complete.

1. Select the template titled *Container security project solution template*.



Container security project solution template
DOCX File

2. Rename the downloaded template with a unique file name so you can easily find it and upload it for submission later. For example, "Container security project solution_Your initials."

3. Select the template titled *Audit logs project solution template*.



Audit logs project solution template
DOCX File

4. Rename the downloaded template with a unique file name so you can easily find it and upload it for submission later. For example, "Audit logs project solution_Your initials."

Guidelines for screenshots

1. Use an application, such as Snipping Tool, to capture screenshots.
2. Perform a step in the task and capture the corresponding screenshot.
3. Confirm that the screenshot displays the result of your activity. You should set your system's display resolution to the highest resolution available.
4. You may also want to mask or remove personal information, such as sign in credentials, from the screenshot.
5. Copy and paste the screenshot below the appropriate step in your version of the relevant deliverable: *Container security project solution* or *Audit logs project solution*.

^ Completing project tasks (1/2): Secure a container with SAS

Creating a well-formatted document with steps and screenshots will help your peer effectively review your submission. Follow the instructions below to create your project deliverable.

Deliverable: Container security project solution

Task: Secure a container with SAS.

Activity 1: Create a Storage account with geo-redundant storage and a container with the public access level set to **Blob**.

You can use the following information while performing this activity:

(**Note:** It is not mandatory that you follow the specific names or settings. However, following the details provided will make it easier when you conduct peer reviews or compare your work with the exemplar you can download at the end of this project.)

- Resource group name: **MCResourcegroup**
- Storage account name: **mcazstorage02**
- Region: **East US**
- Performance: **Standard**
- Container name: **branding**

While performing this activity, document each step and the associated screenshot in your version of the *Container security project solution* document.

Remember to include the following screenshots:

- A screenshot for selecting the **Create** option on the **Storage account** page.
- A screenshot that shows the **Geo-redundant storage (GRS)** option selected on the **Create a storage account** page.
- A screenshot that shows the **Containers** page with the public access level set to **Blob (anonymous read access to blobs only)**.

Activity 2: Upload files to Blob Storage.

You can use the container that you created in the previous activity to perform this activity.

While performing this activity, document each step and the associated screenshot in your version of the *Container security project solution* document.

Remember to include the following screenshots:

- A screenshot that shows the **Containers** page with at least one container listed.
- A screenshot that shows the **Upload blob** pane on the right side.

Activity 3: Create SAS tokens.

You can use the following information while performing this activity:

- Permissions: **Read** and **List**
- Start and end time: Provide access only for a week to align with the problem statement.
- Protocol: **HTTPS**

While performing this activity, document each step and the associated screenshot in your version of the *Container security project solution* document.

Remember to include the following screenshots:

- A screenshot that indicates selecting a container.
- A screenshot that shows the **Generate SAS** pane on the **Containers** page.
- A screenshot that shows a value in **Blob SAS URL**.

Activity 4: Download and install Azure Storage Explorer.

While performing this activity, document each step and the associated screenshot in your version of the *Container security project solution* document.

Remember to include the following screenshot:

- A screenshot of the installation page of Azure Storage Explorer.

Activity 5: Connect to the Storage account using Azure Storage Explorer.

You can use the following information while performing this activity:

- Display name: **Branding**

While performing this activity, document each step and the associated screenshot in your version of the *Container security project solution* document.

Remember to include the following screenshots:

- A screenshot that shows the connection method as **Shared access signature URL (SAS)**.
- A screenshot that shows the connection summary page.
- A screenshot that shows the contents of the container.

^ Completing project tasks (2/2): Configure audit logs

Creating a well-formatted document with steps and screenshots will help your peer effectively review your submission. Follow the instructions below to create your project deliverable.

Deliverable: Audit logs project solution

Task: Configure audit logs.

Activity 1: Create a resource group.

You can use the following information while performing this activity:

(**Note:** It is not mandatory that you follow the specific names or settings. However, following the details provided will make it easier when you conduct peer reviews or compare your work with the exemplar you can download at the end of this project.)

- Resource group name: **AZ500LAB01**
- Region: **East US**

While performing this activity, document each step and the associated screenshot in your version of the *Audit logs project solution* document.

Remember to include the following screenshots:

- A screenshot that shows the **Resource groups** page with the **Create** option.
- A screenshot that shows the **Create a resource group** page with the resource group name.

Activity 2: Create Azure Database for MySQL servers.

You can use the following information while performing this activity:

- Resource group: **AZ500LAB01**
- Server name: **mysqlserverflexible**
- Region: **East US**
- MySQL version: **5.7**
- Workload type: **For development or hobby projects**
- Compute + storage: **Burstable, B1s 1 vCores, 1 GiB RAM, 20 GiB storage, 360 IOPS**

- Availability zone: **No preference**
- High Availability: This should be unchecked.
- Authentication method: **MySQL authentication only**
- Admin username: **mysqlserver**
- Connectivity method: **Public access**

While performing this activity, document each step and the associated screenshot in your version of the *Audit logs project solution* document.

Remember to include the following screenshots:

- A screenshot that shows the **Select Azure Database for MySQL Deployment option** page with the **Flexible server** section.
- A screenshot that shows **MySQL authentication only** as the authentication method.
- A screenshot that shows the **Deployment succeeded** notification for the flexible server.

Activity 3: Create a Key Vault. Then, create a user-assigned managed identity and set permissions for key operations.

You can use the following information while performing this activity:

- Resource group: **AZ500LAB01**
- Key Vault name: **Demo-Keyvault06**
- Region: **East US**
- Pricing tier: **Standard**
- Purge protection: **Enable**
- User-assigned managed identity name: **demo-umi**
- Permissions: **Get, List, Unwrap Key**, and **Wrap Key**

While performing this activity, document each step and the associated screenshot in your version of the *Audit logs project solution* document.

Remember to include the following screenshots:

- A screenshot that shows the **Create a key vault** page with the key vault name entered.
- A screenshot that shows the **Create User Managed Identity** page with the user-managed identity name entered.
- A screenshot that shows the **Create an access policy** page with the **Get, List, Unwrap Key**, and **Wrap Key** permissions listed.

Activity 4: Configure customer-managed key.

While performing this activity, document each step and the associated screenshot in your version of the *Audit logs project solution* document.

Remember to include the following screenshot:

- A screenshot that shows the **Data encryption** page for the flexible server with the Key Vault and key names entered in the **Select a key** pane.

Activity 5: Use data encryption for restore.

You can use the following information while performing this activity:

- Name of the restore server: **restoreserver**

While performing this activity, document each step and the associated screenshot in your version of the *Audit logs project solution* document.

Remember to include the following screenshots:

- A screenshot that shows the **Create Azure Database for MySQL Flexible server – Restore server** page with a name and **Compute + storage** details entered for the restore server.
- A screenshot that shows the **Create Azure Database for MySQL Flexible server – Restore server** page with Key Vault and key details entered in the **Select a key** pane.

Activity 6: Configure an audit log for the connection and general events.

While performing this activity, document each step and the associated screenshot in your version of the *Audit logs project solution* document.

Remember to include the following screenshot:

- A screenshot that shows the **Server parameters** page of the flexible server with **CONNECTION** and **GENERAL** event types selected.

^ Finalizing your assignment

Post completion of all the tasks, you can finalize your deliverables. Ensure that you've added the steps and screenshots for all the tasks.

Remove any instruction sections or examples provided in the original deliverable templates. Your final deliverables should contain only your work. You are then ready to upload your deliverables in the **My submission** tab.

^ Review criteria

Twenty-five (25) points are allotted for this final course project. The points are distributed across the various tasks and activities that you have completed while developing your version of the following deliverables:

- *Container security project solution* (12 pts)
- *Audit logs project solution* (13 pts)

Your grade will be based on the following distribution of points across tasks and activities.

Deliverable 1: Container security project solution

Task: Secure a container with SAS. (12 pts)

Activity 1: Create a Storage account with geo-redundant storage and a container with the public access level set to Blob. (3 pts)

Activity 2: Upload files to Blob Storage. (2 pts)

Activity 3: Create SAS tokens. (3 pts)

Activity 4: Download and install Azure Storage Explorer. (1 pt)

Activity 5: Connect to the Storage account. (3 pts)

Deliverable 2: Audit logs project solution

Task: Configure audit logs. (13 pts)

Activity 1: Create a resource group. (2 pts)

Activity 2: Create an Azure Database for MySQL servers. (3 pts)

Activity 3: Create a Key Vault and a user-assigned managed identity and set the proper permissions for key operations. (3 pts)

Activity 4: Configure a customer-managed key. (1 pt)

Activity 5: Use data encryption for restore. (2 pts)

Activity 6: Configure an audit log for the connection and general events. (2 pts)

^ Preparing for project submission

Do not proceed to the **My submission** and **Peers to review** tabs until you've completed your version of the following deliverables:

- *Container security project solution*
- *Audit logs project solution*

Review, proofread, and edit your deliverables. Ensure they are ready to submit for peer review.

To submit the deliverables, upload your documents to the Coursera platform following the prompts in the **My submission** and **Peers to review** tabs.

^ Peer review

Giving, receiving, and incorporating feedback is a crucial professional skill. After you submit your deliverable, as the next activity, you will have to review and provide feedback on two (2) peer submissions. Similarly,

as the next activity, you will have to review and provide feedback on two (2) peers' submissions. Similarly, your submission will be reviewed by a peer. We've shared a Rubric to ease the review process. You will be presented with questions about your peers' submissions. For example, did your peer document all the key evidence? If prompted, you may have to provide more in-depth, written feedback for some questions. Please provide constructive feedback to help your peers understand what they did well and what they can improve on.

^ Compare your work (Discussion)

After submitting your deliverables and completing two (2) peer reviews, you can access the exemplars: *Container security project solution exemplar* and *Audit logs project solution exemplar*. These exemplars are step-by-step guides for the project tasks. You can compare the exemplars with your submission. Then, reflect on your learning and share your assignment experience with peers in the discussion forum.

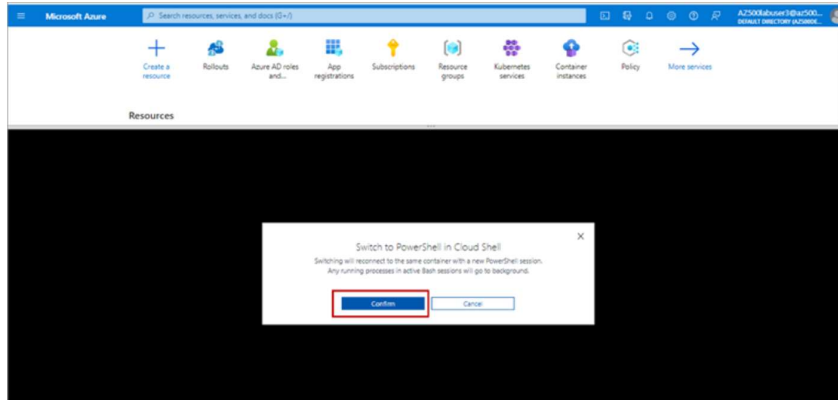
^ After completion of the project

When working on your Azure subscription, at the end of a project, it's good practice to assess whether you still require the resources you created. Running resources can cost you money. You can delete resources individually or delete the entire resource group containing the resources. Here's how you can do this.

Clean up resources

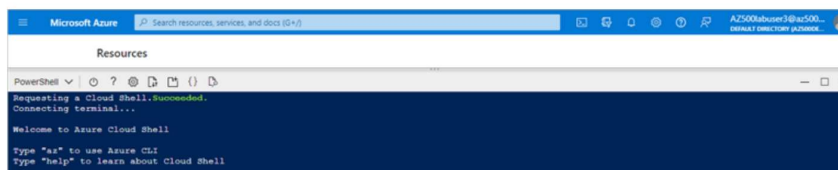
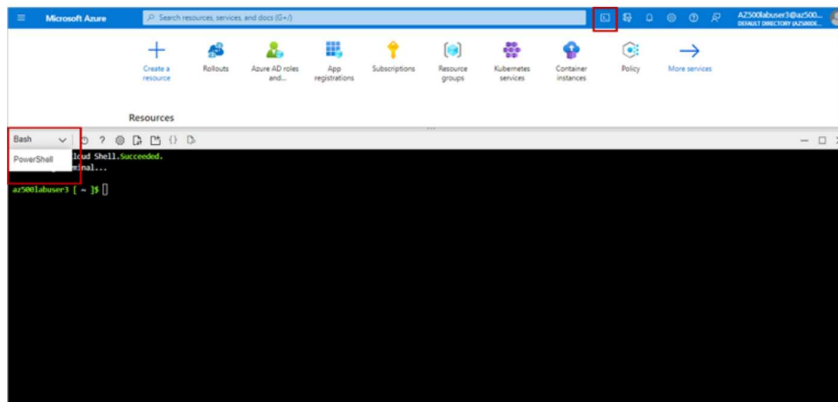
Note: Remember to remove any newly created Azure resources that you no longer use. Removing unused resources ensures you will not incur unexpected costs.

1. In the Azure portal, open the Cloud Shell by clicking the first icon in the top right of the Azure Portal. In the upper-left drop-down menu of the **Cloud Shell** pane, select **PowerShell**.
2. When prompted, select **Confirm**.



3. In the PowerShell session within the **Cloud Shell** pane, run the following code to remove the AZ500LAB01 resource group you might have created for developing the Audit logs project solution:

```
Remove-AzResourceGroup -Name "AZ500LAB01" -Force -AsJob
```



```
NOTED: Azure Cloud Shell now includes Predictive Intelligence! Learn more: https://aka.ms/CloudShell/IntelliSense
VERBOSE: Authenticating to Azure ...
VERBOSE: Building your Azure .dsc
PS /home/az5001abuser3> Remove-AzResourceGroup -Name "AZ5001LAB01" -Force -AsJob
Id      Name      PSJobTypeName State      HasMoreData  Location      Command
--      -
2       Long Running Q. AzureLongRunni.. Running     True         localhost     Remove-AzResourceGroup
PS /home/az5001abuser3>
```

Similarly, run the following code in PowerShell to remove the MCResourcegroup resource group you might have created for developing the Container security project solution document:

```
Remove-AzResourceGroup -Name "MCResourcegroup" -Force -AsJob
```

4. Close the **Cloud Shell** pane.